Aprendizagem 2023/24

Homework I – Group 24

(ist105887, ist106228)

I. Pen-and-paper

1)

First split:

$$H(y_{out} \mid y_2) = \frac{4}{7} H(y_{out} \mid y_2 = 0) + \frac{3}{7} H(y_{out} \mid y_2 = 1)$$

$$= 1.251$$

$$H(y_{out} \mid y_2 = 0) = \frac{1}{2} \log 2 + \frac{1}{4} \log 4 + \frac{1}{4} \log 4 = \frac{3}{2}$$

$$H(y_{out} \mid y_2 = 1) = \frac{2}{3} \log \frac{3}{2} + \frac{1}{3} \log 3 = \boxed{0.918}$$

$$H(y_{out} \mid y_3) = \frac{2}{7} H(y_{out} \mid y_3 = 0) + \frac{4}{7} H(y_{out} \mid y_3 = 1) + \frac{1}{7} H(y_{out} \mid y_3 = 2) = \boxed{0.857}$$

$$H(y_{out} \mid y_3 = 0) = 0 \text{ (unary)}$$

$$H(y_{out} \mid y_3 = 1) = \frac{1}{2} \log 2 + \frac{1}{4} \log 4 + \frac{1}{4} \log 4$$

$$H(y_{out} \mid y_3 = 2) = 0 \text{ (unary)}$$

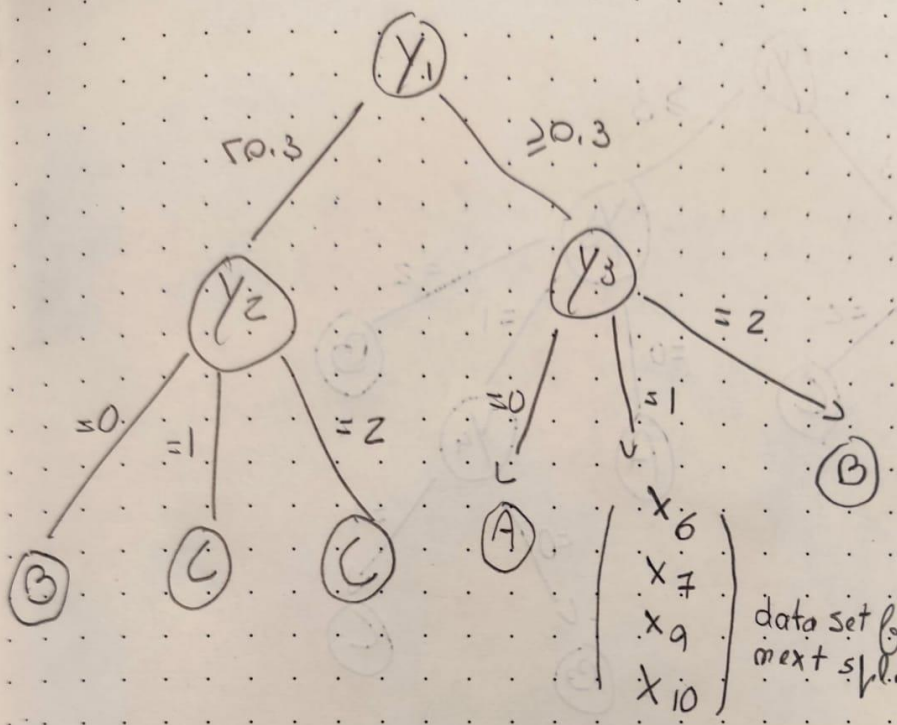
$$H(y_{out} \mid y_4) = \frac{4}{7} H(y_{out} \mid y_4 = 0) + \frac{3}{7} H(y_{out} \mid y_4 = 1) = \boxed{0.965}$$

$$H(y_{out} \mid y_4 = 0) = 1 \text{ (binary)}$$

$$H(y_{out} \mid y_4 = 1) = \frac{2}{3} \log \frac{3}{2} + \frac{1}{3} \log 3$$

$$\boxed{>> H(y_{out} \mid y_3)}$$

Second split:



$H(y_{out} \mid y_2) = H(y_{out} \mid y_2 = 0) = \boxed{\frac{3}{2}}$

$H(y_{out} \mid y_2 = 0) = \frac{1}{2}\log 2 + \frac{1}{4}\log 4 + \frac{1}{4}\log 4$

$H(y_{out} \mid y_4) = \frac{1}{4} H(y_{out} \mid y_4 = 0) + \frac{3}{4} H(y_{out} \mid y_4 = 1)$

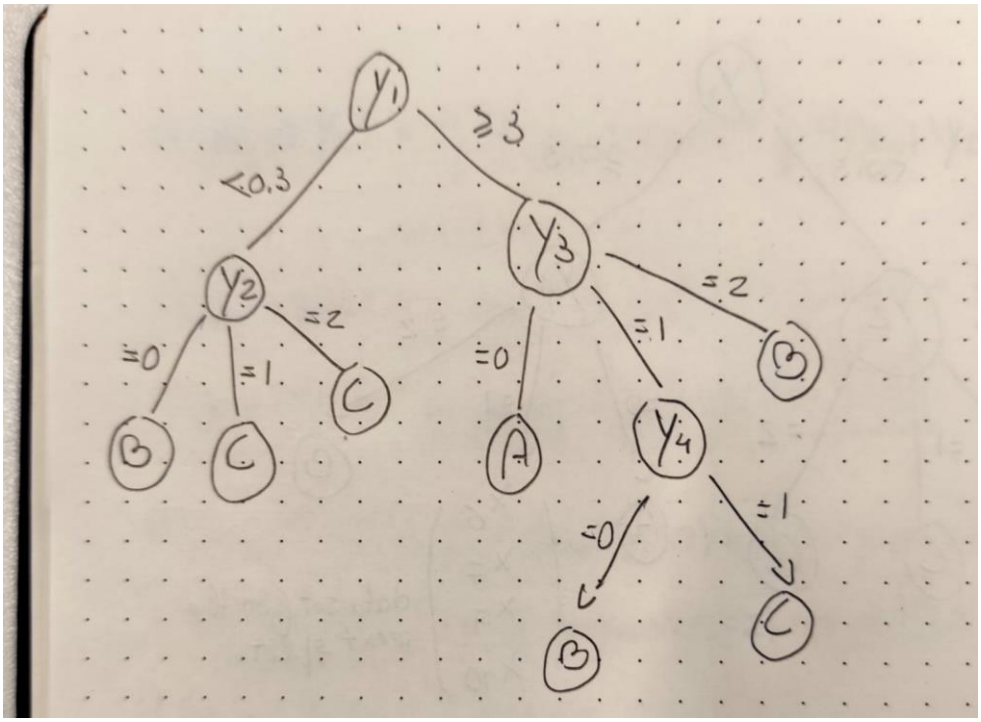
$= \boxed{0.689}$

$H(y_{out} \mid y_4 = 0) = 0$ (unary)

$H(y_{out} \mid y_4 = 1) = \frac{1}{3}\log 3 + \frac{2}{3}\log\frac{3}{2}$

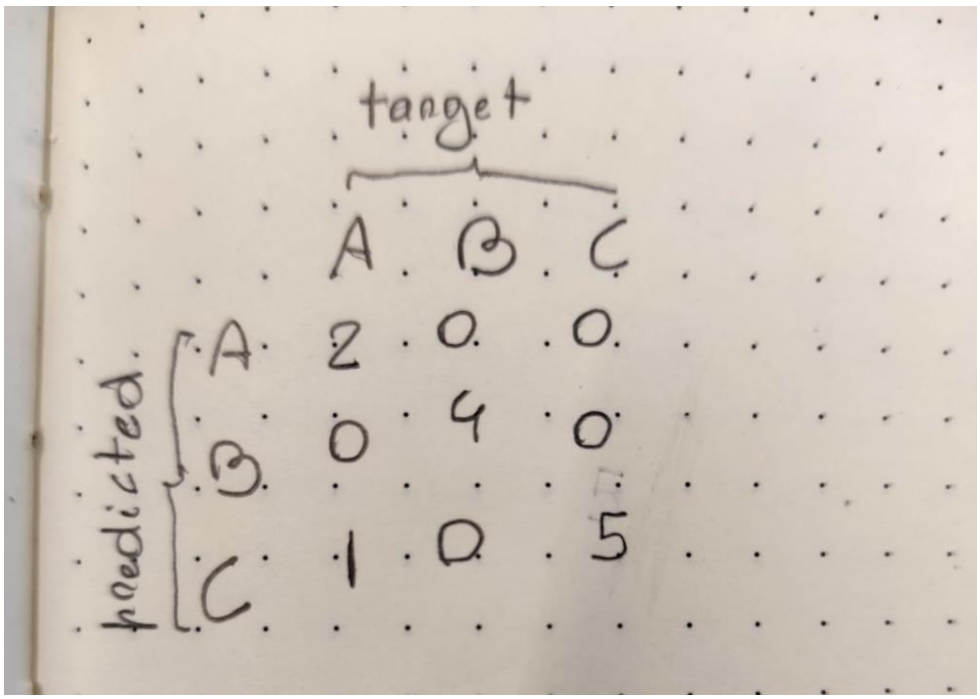
$\boxed{> H(y_{out} \mid y_4)}$

Decision tree:



2)

Confusion matrix:



predicted \ target	A	B	C
A	2	0	0
B	0	4	0
C	1	0	5

3)

First table is a counter of True Positives, True Negatives, False Negatives and False Positives per class, respectively.

Second table is the precision and recall per class, where precision is equal to the positive observations predicted as positive (TP/(TP+FN)) and recall is equal to the positive observations among the observations predicted as positive (TP/(TP+FP)).

Third table is the balanced F1 score per class (2PR/(P+R)).

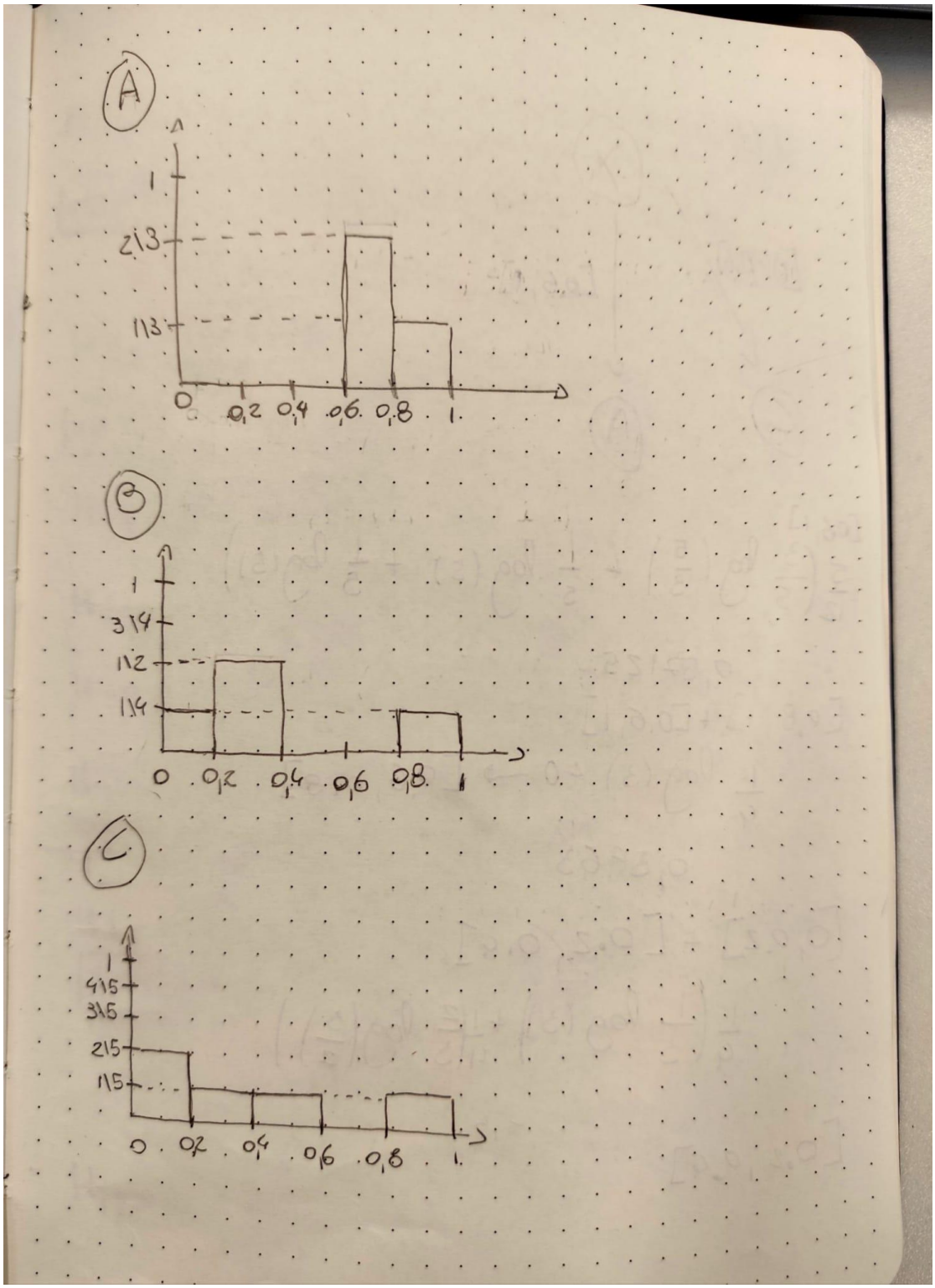
	TP	TN	FN	FP
A	2	9	0	1
B	4	8	0	0
C	5	6	1	0

	Precision	Recall
A	66.6%	100%
B	100%	100%
C	100%	83.3%

	F1 SCORE
A	0.8
B	1
C	0.91

4)

Class-conditional relative histograms:



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To find the n-ary root split it was calculated the IG for each bin and for some bin conjunctions that shared the same target class. However, the IG proved to be higher for the individual intervals, which means that a decision tree with a 5-ary root split (one for each bin) would be optimal.

$$H(y_1 \in [0, 0.2]) = \underbrace{\frac{1}{4}}_{P(y_1 \in [0,0.2])} \left(0 + \underbrace{\frac{1}{3}\log(3)}_{\substack{\text{class B} \\ y_1 \in [0,0.2]}} + \underbrace{\frac{2}{3}\log\left(\frac{3}{2}\right)}_{\substack{\text{class C} \\ y_1 \in [0,0.2]}} \right) \boxed{= 0.230}$$

$$H(y_1 \in [0.2, 0.4]) = \underbrace{\frac{1}{4}}_{P(y_1 \in [0.2,0.4])} \left(0 + \underbrace{\frac{2}{3}\log\left(\frac{3}{2}\right)}_{\substack{\text{class B} \\ y_1 \in [0.2,0.4]}} + \underbrace{\frac{1}{3}\log(3)}_{\substack{\text{class C} \\ y_1 \in [0.2,0.4]}} \right) \boxed{= 0.230}$$

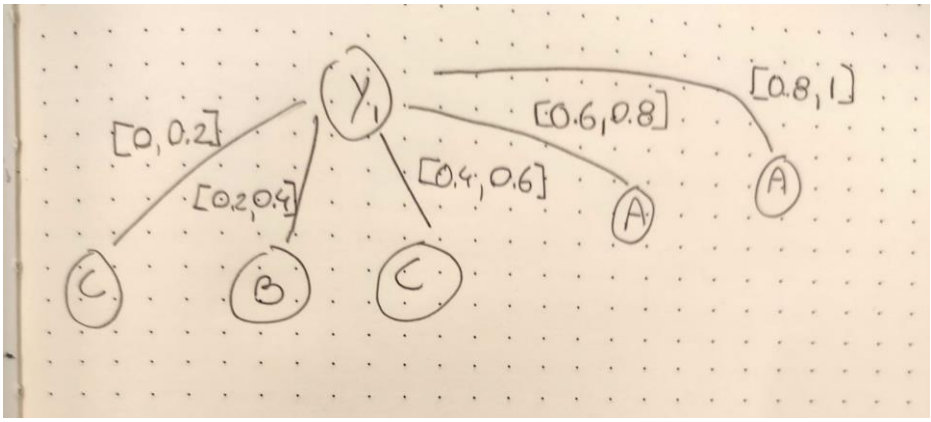
$$H(y_1 \in [0.4, 0.6]) \boxed{= 0}$$

$$H(y_1 \in [0.6, 0.8]) \boxed{= 0}$$

$$H(y_1 \in [0.8, 1]) = \frac{1}{4}\log(3) \boxed{= 0.346}$$

$$H(y_1 \in [0, 0.2] \cup [0.4, 0.6]) = \frac{1}{3}\left(\frac{1}{4}\log 4 + \frac{3}{4}\log\frac{4}{3}\right) \boxed{= 0.270}$$

$$H(y_1 \in [0.6, 1]) = \frac{5}{12}\left(\frac{3}{5}\log\left(\frac{5}{3}\right) + \frac{2}{5}\log(5)\right) \boxed{= 0.571}$$

$$H(y_1 \in [0, 0.2]) + H(y_1 \in [0.2, 0.4]) < H(y_1 \in [0, 0.2] \cup [0.2, 0.4])$$

$$H(y_1 \in [0.6, 0.8]) + H(y_1 \in [0.8, 1]) < H(y_1 \in [0.6, 1])$$